

Drug tolerance takes shape: persistent homology reveals cyclic cell-state plasticity in EGFR-mutant lung cancer

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Abstract

Drug-tolerant persister cells drive relapse after EGFR-targeted therapy in lung cancer, yet current single-cell methods cannot quantify the reversible, cyclic state transitions that produce them. Here we apply topological data analysis, specifically persistent homology and the Mapper algorithm, to longitudinal single-cell RNA-seq data from five EGFR-mutant lung-cancer systems spanning cell lines, patient-derived xenografts, and patient tumours. We show that a topological measure of cyclic cell-state structure increases monotonically with drug exposure across all systems, survives rigorous controls for cell-cycle confounding and batch effects, and replicates in a 14-patient longitudinal cohort. Loop-associated genes are enriched for epithelial–mesenchymal transition programs, with significant overlap between cell-line and xenograft models. Our results establish cyclic topological plasticity as a reproducible, cross-scale phenotype of drug tolerance and provide an open-source framework for detecting it in other perturbation contexts.

Keywords: topological data analysis; persistent homology; Mapper; single-cell RNA-seq; drug tolerance; EGFR-mutant lung cancer; cell-state plasticity; computational framework.

1 Introduction

First-line osimertinib has transformed care for EGFR-mutant non-small cell lung cancer (NSCLC), yet essentially every patient eventually relapses from residual disease [1, 2]. The residual cells,

known as drug-tolerant persisters, survive targeted therapy without acquiring resistance mutations [3, 4]. Early work described persisters as a single quiescent state, but recent lineage-tracing experiments have revealed that individual cells reversibly transition among multiple transcriptional programs before either dying or acquiring stable resistance [5, 6]. This inter-state cycling is distinct from one-directional progression along a differentiation hierarchy and is thought to underlie the adaptive plasticity that makes persisters so difficult to eradicate.

Standard single-cell trajectory tools, including branching-graph methods such as Monocle and Slingshot, connectivity-graph methods such as PAGA, and vector-field methods such as RNA velocity, were designed to model one-directional progressions and therefore cannot quantify how robustly cells trace a closed loop in state space [7]. When drug-tolerant cells reversibly cycle among states, each cell is a point in gene-expression space and the collective motion traces a loop: the same kind of closed one-dimensional structure that distinguishes a donut from a sphere in topology (fig. 1a–b). Persistent homology, a branch of computational topology, measures the robustness of such loops across spatial scales, and the Mapper algorithm [8] complements this with an interpretable graph summary. Both have been applied to bulk cancer transcriptomes [9] and developmental single-cell data [10], but have not been integrated into a reproducible framework for longitudinal perturbation studies with formal statistical testing.

Here we present a framework that applies persistent homology and Mapper to longitudinal single-cell RNA-seq data, with integrated permutation-based null testing, cell-cycle controls, and Mapper-based gene attribution. We validate the approach on five independent EGFR-mutant lung-cancer datasets spanning two drugs (erlotinib, osimertinib), three biological scales (cell line, patient-derived xenograft, patient tumour), and a 14-patient longitudinal cohort. We show that a topological measure of cyclic cell-state structure, the maximum H_1 persistence, increases monotonically with drug exposure across all systems, is not explained by cell-cycle variation or batch effects, and is biologically anchored by enrichment of epithelial–mesenchymal transition (EMT) programs in loop-associated transcripts. All analyses are reproducible from raw data via a single command under an open-source license.

2 Results

Cyclic topology in erlotinib-treated PC9 cells

We first asked whether drug-treated NSCLC cells exhibit cyclic topological structure. We analysed the PC9 erlotinib time-series (GSE134839 [6]; six timepoints D0–D11; fig. 2; Methods). All drug-treated timepoints showed non-zero H_1 persistence, with maximum values ranging from 1.08 (D11) to 1.77 (D9) (fig. 3, table 1). Pairwise Wasserstein distances between persistence diagrams separated D0 from all treated timepoints, indicating that the distribution of topological features differs most strongly between untreated and treated conditions (fig. 3). Per-timepoint sample sizes (200–379 cells) limited individual permutation-test power; the larger validation cohorts below recover significance (table 2).

Osimertinib drives a monotonic topological increase across cell line and patient-derived xenografts

To test whether the cyclic signal strengthens with drug exposure, we analysed a PC9 osimertinib time-series with Watermelon lineage tracing (GSE150949 [5]; D0/D3/D7/D14; Methods). Max H_1 rose monotonically from 1.41 at D0 to 3.78 at D14 (2.7-fold; fig. 4), with permutation tests confirming $p < 0.01$ at D7 and D14 (table 2).

To determine whether this finding extends beyond cell culture, we analysed two EGFR-mutant PDX models (YU-003, YU-006) under matched untreated and osimertinib-residual conditions (GSE243562 [11]; Methods). Both models showed increased max H_1 under residual disease: YU-003 from 2.79 to 3.85 (1.4-fold, $p < 0.01$); YU-006 from 2.81 to 6.08 (2.2-fold, $p < 0.01$) (fig. 4, table 2). This replicates the cell-line finding in an *in vivo* system with stromal, vascular, and host-immune context.

A timepoint-label permutation test on osimertinib D14 vs. D0 further confirmed that the observed topological change depends on drug exposure rather than random point-cloud structure ($p < 0.02$; Methods).

The topological signal is not explained by cell-cycle variation or batch effects

Cell-cycle oscillation produces a cyclic snapshot signature that could confound the topological signal. We applied two independent controls. First, we removed cell-cycle genes from the feature

set and recomputed persistence; the ratio of max H_1 after to before removal ranged from 0.71 to 1.11 across all six timepoints (fig. 3, table 3). Second, we regressed out S-phase and G2/M-phase scores from the expression matrix; this *increased* max H_1 in all five tested cohorts (post/pre ratio 1.07–4.05; table 3). Together, these results indicate that the observed cyclic topology is not driven by cell-cycle variation (Methods).

To rule out batch or platform effects, we recomputed max H_1 on Harmony-corrected embeddings [12]. The osimertinib monotonic trend, the YU-006 untreated-to-residual difference, and the patient-tumour baseline all survived correction (fig. 6; Supplementary Table S10), indicating that the topological signal reflects biology rather than technical artefacts.

Clinical validation in a 14-patient longitudinal cohort

To establish a topological reference for human lung adenocarcinoma, we analysed a treatment-naive patient LUAD atlas (GSE131907 [13]; Methods). Treatment-naive patient tumours showed max $H_1 = 4.77$, substantially higher than treatment-naive cell line (1.41) or PDX (2.79, 2.81) baselines (fig. 5). Drug treatment moves cell-line and PDX topology from the baseline regime toward the patient-tumour range (fig. 5; the within-system monotonic increase is the primary claim, as cross-system comparisons are confounded by platform and cell-count differences).

To directly test the framework on clinical tissue, we analysed the EGFR-mutant subset of Maynard et al. [14] (14 patients with biopsies at treatment-naive, persister, and progressive disease stages; Methods). Pooled max H_1 grew monotonically across the three stages: 5.37 (treatment-naive) $\rightarrow$ 7.13 (persister) $\rightarrow$ 8.05 (progressive disease) (1.5-fold; fig. 7a). Three patients had matched pre/post-treatment biopsies; max H_1 increased in all three (median 1.6-fold, range 1.35–3.25-fold; Supplementary Table S11), supporting the use of max H_1 as a within-patient relative-change readout.

EMT programs underlie cyclic topology

To identify the transcriptional programs underlying loop formation, we constructed Mapper graphs and computed gene-level connectivity scores (Methods). In the cell-line analysis, top-connected genes were canonical EMT markers (*TPM1*, *KRT8*, *KRT19*); MSigDB Hallmark EMT ranked first among enriched pathways. The PDX analysis recovered a broader mesenchymal/secretory signature (*S100A4*, *LGALS3*, *COL1A1*), consistent with an EMT-like plastic state realised

through a partially shared repertoire. Eleven genes overlap between the two top-100 lists (hypergeometric $p = 1.25 \times 10^{-11}$), indicating significant but incomplete overlap between cell-line and PDX plasticity programs. Differential expression recovered EMT-related signatures in both systems, with system-specific secondary pathways (Supplementary Tables S2, S2', and S4).

Using an unbiased PC1 filter in the Mapper, cell-cycle markers co-ranked with EMT markers, indicating that the loops reflect a multi-program state space rather than a pure EMT signature (Supplementary Table S2'). Variance-normalised connectivity scores strengthened the EMT enrichment, confirming it is not an expression-magnitude artefact (Supplementary Table S9).

A generalisable, reproducible framework

The pipeline comprises fifteen independent modules coordinated by a single command (Methods). It ran without modification on Drop-seq and 10x cell-line data, 10x PDX data, and a 208,506-cell patient atlas. Applying it to a new dataset requires only a processed expression matrix with sample annotations and a choice of Mapper filter (PC1, EMT score, or diffusion component; Methods). Resampling stability, replication under alternative persistence summaries, and Mapper parameter robustness are reported in Supplementary Methods S10–S12.

3 Discussion

We have shown that cyclic topological structure in single-cell transcriptomes increases monotonically with EGFR-TKI exposure across five independent systems spanning cell lines, patient-derived xenografts, and patient tumours. The primary contribution is a reproducible, open-source framework that reframes the question from whether drug tolerance produces topological cyclicity in any single dataset to whether topological cyclicity is a reproducible cross-system phenotype. Our answer is yes: $\max H_1$ increases with drug exposure in every within-system comparison, all six validation-cohort permutation tests reach $p < 0.05$, and the signal survives cell-cycle controls and batch correction.

Our framework complements lineage-tracing studies that established cycling subpopulations in PC9 [5] and identified discrete DTP states [6] by detecting cyclic behaviour directly from scRNA-seq without lineage-tracing infrastructure. Relative to earlier applications of Mapper to developmental scRNA-seq [10], we add permutation-based null testing, cell-cycle controls, and application to perturbation-response data. The broader non-genetic drug-tolerance literature,

including YAP-mediated senescent persisters [15], melanoma minimal-residual-disease plasticity [16], and non-genetic resistance mechanisms [17, 18], provides biological context. Our contribution is a quantitative, topologically grounded statistic for detecting and comparing such structures.

An informative finding is that the topological signature replicates across cell line and PDX with higher fidelity than the underlying gene programs. The gene overlap between systems is highly significant but incomplete, suggesting that plasticity is realised through partially different repertoires depending on microenvironment, consistent with context-dependent EMT programs [19]. The topological statistic abstracts over these specific gene choices, making it a more stable cross-system readout than any individual gene signature.

If drug tolerance operates through topologically cyclic plasticity, therapeutic combinations that collapse the cycle may outperform single-state-targeting strategies. The bidirectional nature of inter-state cycling further provides a conceptual basis for the partial clinical efficacy of TKI rechallenge after a drug holiday [20, 21]: cells that survived prior treatment can revert to drug-sensitive states. The cross-scale reproducibility of the signature argues that $\max H_1$ is suitable for benchmarking drug responses across labs and may generalise to other perturbation contexts such as immunotherapy escape or chemotherapy-induced dedifferentiation.

Several limitations should be noted. Cross-system comparisons mix platforms (Drop-seq and 10x Chromium); the within-system monotonic increase is therefore the primary claim. The three matched patient biopsies are hypothesis-generating rather than confirmatory; a prospective paired-biopsy study would be needed for within-patient validation. Cell-cycle gene ablation removes a standard but not exhaustive gene set; the complementary S/G2M regression control addresses this limitation. A full cross-platform integration using methods such as scVI [22] is deferred to future work.

We speculate that cyclic plasticity reflects a Waddington-style limit cycle around a shallow epigenetic basin [23, 24], making it adaptive as bet-hedging under fluctuating drug selection [25, 26]. Distinguishing selection of pre-existing cycles from drug-induced de novo cycles would require lineage tracing with controlled clonal bottlenecks. Adaptive therapy and state-trapping combinations are testable hypotheses against the topological readout established here.

4 Conclusions

Drug treatment does not simply push cancer cells along a one-way path to resistance; it reshapes the cell-state landscape into a form that admits reversible inter-state cycling. The topological framework presented here quantifies this cyclic plasticity and reveals that it increases monotonically with drug exposure across cell lines, xenografts, and patient tumours. The approach generalises to any setting where cells reversibly explore multiple transcriptional states, a phenomenon that standard trajectory methods are designed to ignore.

5 Methods

Datasets

All cell models harbour activating *EGFR* mutations sensitising them to EGFR-TKIs; PC9 carries the canonical exon 19 deletion E746-A750del. Per-cohort details: GSE134839 [6]: PC9 erlotinib time-series; six Drop-seq DGE matrices at 2 μ M (D0, D1, D2, D4, D9, D11); 6,508 total cells, 2,942 post-QC. GSE150949 [5]: PC9 with Watermelon lineage tracing under osimertinib; 10x Chromium; 56,419 cells across D0/D3/D7/D14; subsampled to 5,000 (4,999 post-QC). GSE243562 [11]: two EGFR-mutant PDX models (YU-003, YU-006) under matched untreated and osimertinib-residual conditions; 10x Chromium; 22,032 cells (18,874 post-QC) across four samples, subsampled to 1,500 per group for VR computation. GSE131907 [13]: treatment-naive patient LUAD atlas (mixed driver genotypes); 10x Chromium; 208,506 cells across 44 patients; subsampled to 5,000 tLung epithelial cells, then 1,500 for tractable Vietoris–Rips computation (filter and seed details in Supplementary Methods S1). Maynard 2020 [14]: EGFR-mutant NSCLC longitudinal cohort; Smart-seq2; 13,244 epithelial cells across 14 patients with biopsies at three treatment stages (treatment-naive TN, 3,287 cells; persister/residual disease PER, 4,893 cells; progressive disease PD, 5,064 cells); 1,500-cell subsamples per stage.

Preprocessing

We used scanpy v1.11 [27] with standard QC (`min_genes = 200`, `max_genes = 5000`, `max_pct_mito = 20%`, `min_cells = 3`), total-count normalisation to 10,000 counts per cell, log1p transformation, 3,000 highly-variable genes, regression of total counts and mitochondrial percentage, scaling to max 10, and PCA with 50 components (`random_state = 42`). HVG batch keys per cohort:

`timepoint` for GSE134839 and GSE150949, `pdx` for GSE243562, and `Sample` for GSE131907 (rationale in Supplementary Methods S2). For PDX data, mouse genes were removed by a symbol-pattern heuristic (Supplementary Methods S3; 29% of features filtered; limitations in Discussion). Cell-cycle scores were computed from Tirosh et al. [28] S- and G2/M-phase gene lists via `sc.tl.score_genes_cell_cycle`.

Persistent homology

We used `ripser` 0.6 [29, 30] to compute the Vietoris–Rips filtration up to H_1 with the Euclidean metric on the top-30 PCA coordinates; homology is computed over $\mathbb{F}_2$ coefficients (ripser default). Maximum persistence is defined as $\max(\text{death} - \text{birth})$ over finite H_1 bars; this is a hybrid statistic whose existence is a topological invariant but whose numerical scale is metric-dependent (Supplementary Methods S4). Wasserstein distances between diagrams were computed with `persim` 0.3 using the 2-Wasserstein metric. The p -Wasserstein distance between persistence diagrams is stable under bounded perturbations of the input point cloud given a finite-degree- p total-persistence hypothesis [31, 32].

Principal component count sensitivity

Max H_1 depends on the number of PCs retained: across $n_{\text{PCs}} \in \{10, 20, 30, 50\}$ absolute values vary within a factor of two, but the drug-treated $>$ untreated ordering is preserved at every PC count. The $n_{\text{PCs}} = 30$ default follows prior scTDA work [10]. Full sweep numbers and the rationale for the hybrid-statistic framing are in Supplementary Methods S4–S5 and Supplementary Table S5.

Permutation tests

Two complementary null models were used. For the *gene-label null*, we shuffled expression values within each cell (permuting gene labels independently for every cell), re-fit PCA using `sklearn`, and recomputed max H_1 persistence. The p -value is the fraction of null permutations with max $H_1 \geq$ the observed value. Discovery cohort used 500 permutations per timepoint; validation cohorts used 100 permutations on 1,000-cell subsamples for tractability.

For the *timepoint-label null*, we pooled cells from osimertinib D0 and D14 ($n = 2,499$), shuffled the drug-exposure annotation across cells, and recomputed H_1 on the pooled set in each permutation. Over 50 permutations, the observed $\Delta H_1 = 2.07$ exceeded the 95th percentile of the null (1.18) in

0/50 trials, giving $p < 0.02$ (Phipson–Smyth +1 correction). This test directly assesses whether H_1 depends on drug exposure rather than on non-random point-cloud structure alone.

Cell-cycle controls

Two independent controls were applied. For *Tirosh cell-cycle gene ablation*, we removed 39 of 97 Tirosh et al. cell-cycle genes [28] present in the HVG set, re-scaled the remaining features, and recomputed PCA and persistent homology. The ratio of max H_1 after to before ablation ranged from 0.71 to 1.11 across all six timepoints (mean 0.90 ± 0.14); D11 actually increased after cycle-gene removal (ratio = 1.11). The osimertinib cohort additionally contains zero Tirosh genes in its HVG set, ruling out Tirosh-specific contribution to that cohort by absence.

For *S/G2M score regression*, we applied `sc.pp.regress_out` on `S_score` and `G2M_score` for 1,000-cell subsamples across five cohorts. Post/pre ratios ranged from 1.07 to 4.05 (mean 1.97), with every value exceeding 1, confirming that the H_1 signal is preserved after removing the dominant cell-cycle variance axis. Full ratios are reported in table 3.

Harmony batch correction

To assess sensitivity to batch effects, we applied Harmony [12] to the top-50 PCA components with dataset-appropriate batch keys (full parameters in Supplementary Methods S7). Max H_1 was recomputed on the Harmony-corrected embeddings. The osimertinib monotonic trend survived (1.35 $\rightarrow$ 3.94; 2.9-fold, vs. 1.71 $\rightarrow$ 3.78 raw); the YU-006 untreated $\rightarrow$ residual difference was preserved (2.59 $\rightarrow$ 5.19 Harmony vs. 2.36 $\rightarrow$ 5.01 raw); YU-003 was directionally consistent; the Kim LUAD naive baseline was robust (Harmony 4.52 vs. raw 4.77). Full values: Supplementary Table S10.

Mapper

We used `kmapper` 2.0 [33] with an auto-tuned DBSCAN clusterer (`eps` set to the 50th percentile of 10th-nearest-neighbor distances in each cover element). Default parameters were $n_{\text{cubes}} = 15$ and `perc_overlap` = 0.3. Parameter sensitivity was assessed across $n_{\text{cubes}} \in \{10, 15, 20, 25, 30\}$ and `overlap` $\in \{0.2, 0.3, 0.4, 0.5\}$.

EMT score (used as Mapper filter for the cell-line analysis) was computed via `scanpy.tl.score_genes` on a 33-gene MSigDB Hallmark EMT cassette (28 of 33 present in GSE134839; full gene list in

253 Supplementary Methods S6).

254 **Gene connectivity**

255 For each gene g and Mapper node N with cells C_N , we computed $|\text{mean}(g \mid C_N) - \text{mean}(g \mid$
256 all cells)| $\times |C_N|$, summed over all nodes, and divided by total cells. Higher scores indicate genes
257 that vary more between Mapper nodes than globally.

258 **Pathway enrichment**

259 Gene set enrichment was performed with `gseapy 1.1` against `MSigDB_Hallmark_2020` and
260 `GO_Biological_Process_2021`.

261 **Computational environment**

262 All analyses ran on Python 3.11 in a conda environment specified in `envs/environment.yml`;
263 random seed 42 is used for every stochastic operation. Full package versions and platform notes
264 (macOS arm64 and Linux x86_64): Supplementary Methods S14.

265 **Pipeline architecture and unit tests**

266 The framework is organised as fifteen numbered modules (`01_preprocess.py` through `15_maynard_full_cohort`
267 that each accept a standard `AnnData` object and produce typed intermediate outputs. Users
268 can substitute their own datasets by providing an `AnnData` file with a `timepoint` observation
269 column. Full module list and the `pytest` unit-test suite (8 tests, all passing) are described in
270 Supplementary Methods S8–S9.

271 **Code availability**

272 Code is available at <https://github.com/htlin222/sctda-cancer-plasticity> (MIT license).
273 The version-tagged software release accompanying this submission is `v2.7.1-submission`,
274 archived as a public GitHub release at [https://github.com/htlin222/sctda-cancer-plasticity/](https://github.com/htlin222/sctda-cancer-plasticity/releases/tag/v2.7.1-submission)
275 [releases/tag/v2.7.1-submission](https://github.com/htlin222/sctda-cancer-plasticity/releases/tag/v2.7.1-submission). The complete pipeline is reproducible via `make pipeline`
276 from raw GEO data to all figures and tables. A unified `sctda` CLI is planned for v2.0; the current
277 pipeline is invoked via the Makefile and individual `scripts/OX_*.py` files. A `CITATION.cff` file
278 at the repository root provides machine-readable citation metadata for software-citation tooling.

Data availability

All raw scRNA-seq data used in this study are publicly available from the Gene Expression Omnibus under accessions GSE134839, GSE150949, GSE243562, and GSE131907; the Maynard et al. 2020 EGFR-mutant cohort was obtained from the authors' public Google Drive archive referenced in Maynard et al. [14]. Processed AnnData objects, persistence diagrams, and Mapper graphs are regenerable end-to-end from the raw GEO data by running `make pipeline`; seeds and parameters are pinned in the repository so derived outputs are bit-stable across runs.

Declarations

Ethics approval and consent to participate

Not applicable; all analyses use previously published, de-identified public datasets.

Consent for publication

Not applicable.

Availability of data and materials

All raw data and processed intermediate files are available as described in the Methods section (Data availability). Supplementary tables accompanying this manuscript: S1 (cell-cycle ablation permutation p -values), S2 (EMT-filter Mapper gene connectivity), S2' (PC1-filter Mapper gene connectivity), S3 (Mapper parameter sweep), S4 (PDX gene attribution), S5 (Max H_1 vs principal component count), S6 (functional persistence summaries: max H_1 , total persistence, persistence entropy across ten cohorts), S7 (subsample variance: mean $\pm$ SD of max H_1 across five random seeds for four cohorts), S8 (Mapper parameter-sweep Spearman rank correlations), S9 (variance-normalized Mapper gene connectivity), S10 (Harmony batch-correction raw vs. corrected max H_1 comparison), S11 (Maynard 2020 full EGFR-mutant cohort per-patient max H_1 and within-patient trajectory statistics (14 patients, 3 matched pre/post pairs)), S-stringent-CC (stringent cell-cycle regression ratios; reproduced in the lower block of Table 3).

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

H.-T.L. (first author) and Y.-H.T. (corresponding author) jointly conceived the study and designed the analytical strategy. H.-T.L. implemented the pipeline, performed all analyses, produced all figures, and wrote the original draft (CRediT: Conceptualisation, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Visualisation). Y.-H.T. supervised the project, contributed methodological and validation-strategy review, and revised the manuscript (CRediT: Conceptualisation, Methodology, Supervision, Writing – review & editing, Project administration). Both authors read and approved the final manuscript.

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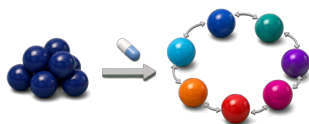
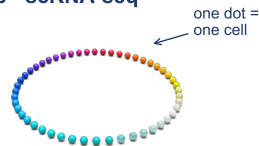
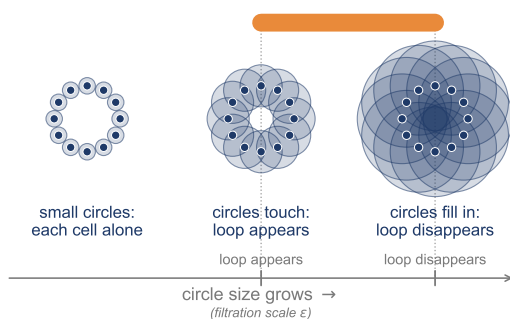
Drug-induced cell-state cycling, detected as a topological loop and measured by persistent homology*(a) drug pressure induces cycling → (b) scRNA-seq sees it as a loop → (c) persistent homology measures the loop's lifespan***a · biology***Drug pressure makes cancer cells cycle between states***b · scRNA-seq***scRNA-seq sees the cycle as a closed loop**quantify with persistent homology***c · how we measure the loop****How long the loop survives**
(max H_1 persistence)

Figure 1: Drug-induced cell-state cycling, detected as a topological loop and measured by persistent homology. (a) Untreated cancer cells occupy a uniform transcriptional state; under EGFR-TKI pressure they diversify into multiple states with reversible transitions between them (cells coloured by state). (b) In scRNA-seq, this reversible cycling appears as a closed loop in cell-state space (one dot = one cell). (c) Persistent homology measures the loop. Around each cell we grow a ball; as the ball radius (the filtration scale ε) increases, the loop is born when balls touch their neighbours and dies when balls fill the loop's interior. The lifespan of this loop, the lifetime of the longest H_1 bar, is the *maximum H_1 persistence* statistic used throughout this work. The mug-and-donut topological-equivalence analogy that motivates the use of homology is described in the Background. Cross-cohort fold-changes are in fig. 5.

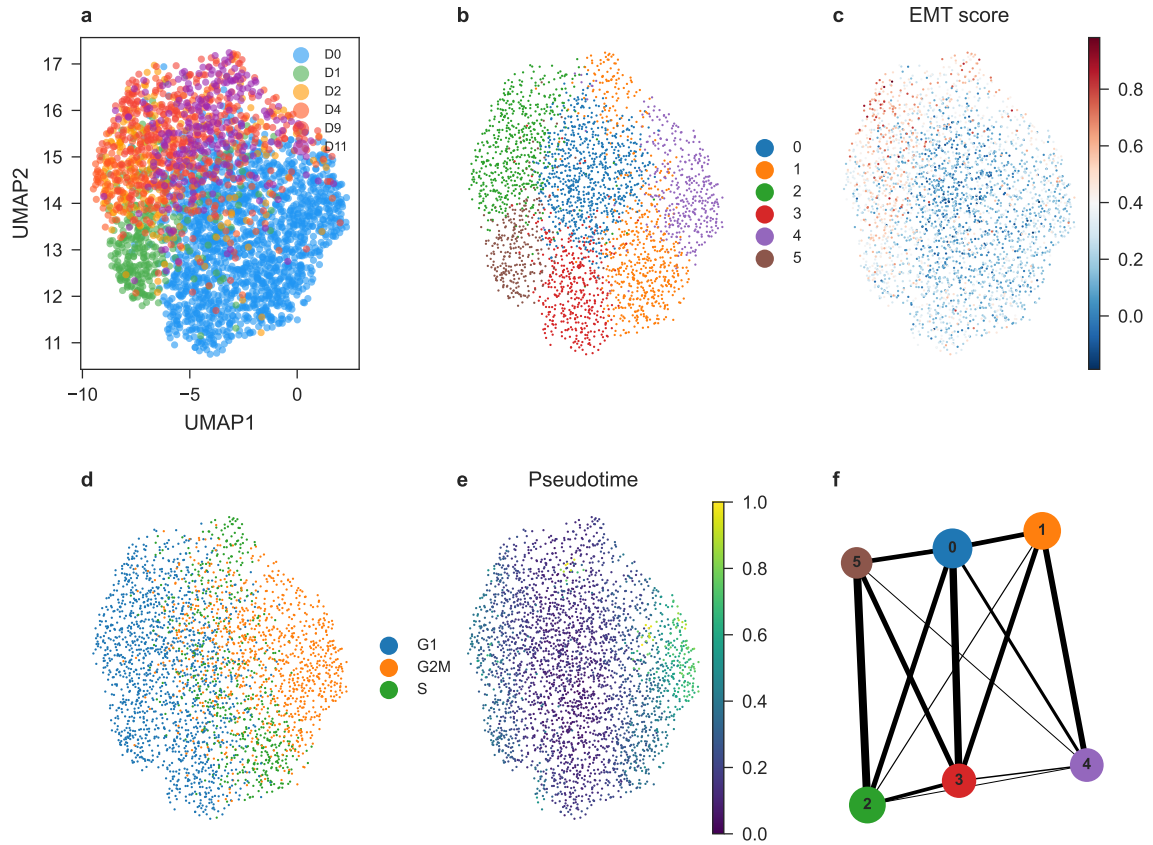
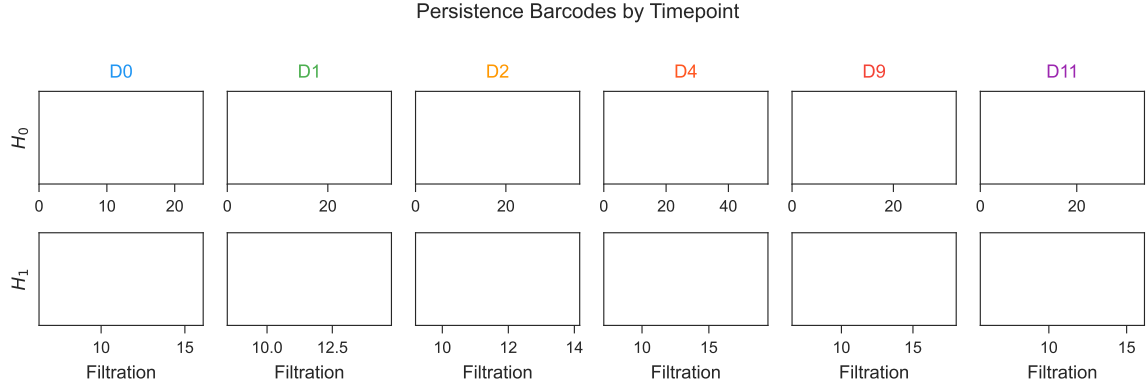
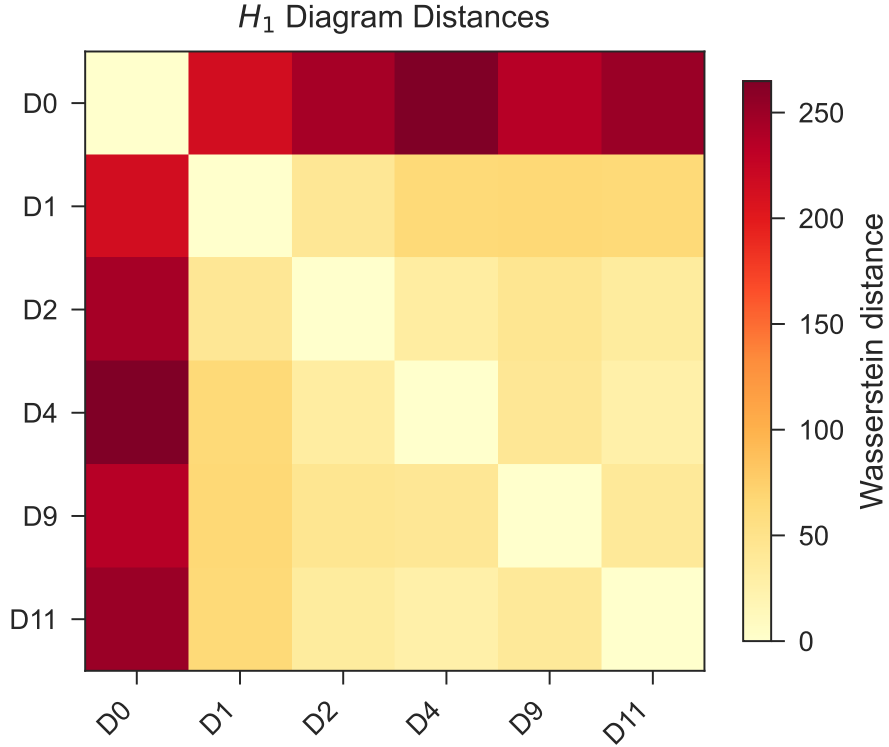


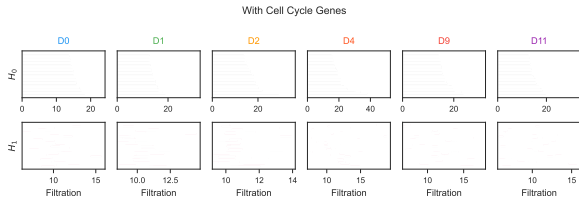
Figure 2: **Study design and cell-line baseline analysis.** (a) UMAP of PC9 erlotinib time course colored by timepoint. (b) Leiden clusters. (c) EMT score. (d) Cell cycle phase. (e) Diffusion pseudotime. (f) PAGA graph. $n = 2,942$ cells across six timepoints.



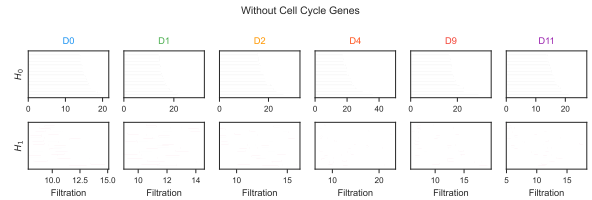
(a) Persistence barcodes by timepoint (top: H_0 ; bottom: H_1).



(b) Pairwise Wasserstein distances between per-timepoint H_1 persistence diagrams.

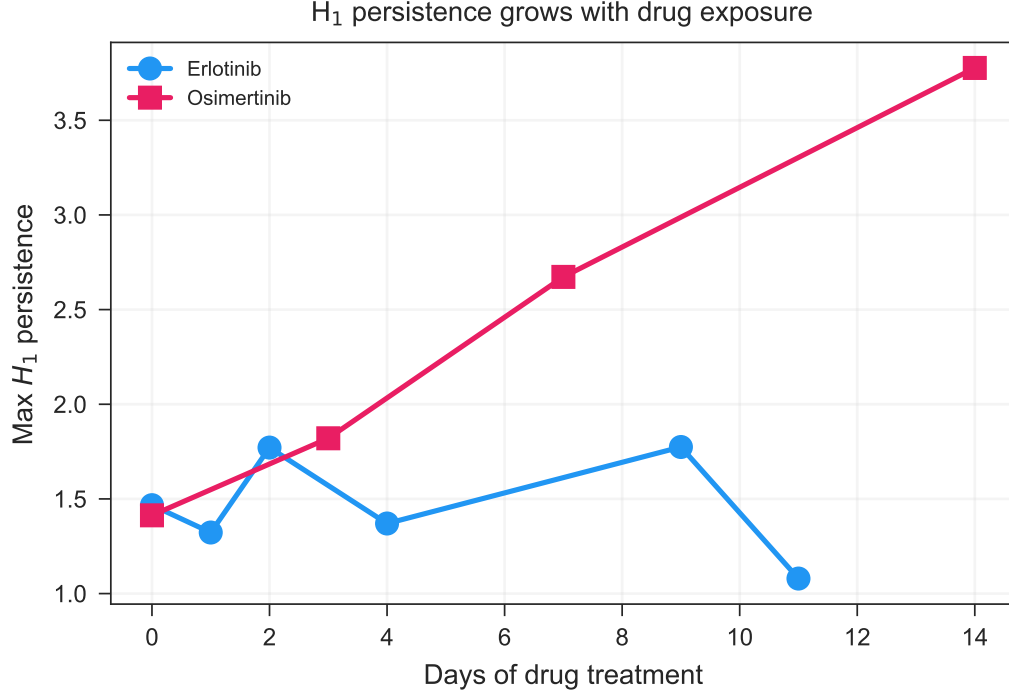


(c) Barcodes with cell-cycle genes.

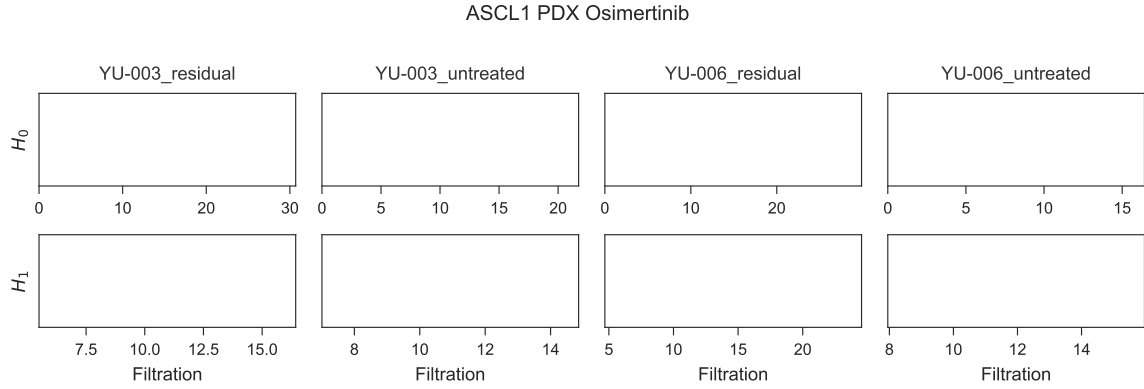


(d) Barcodes without cell-cycle genes.

Figure 3: Persistent homology of the PC9 erlotinib discovery cohort and Tirosh cell-cycle control. (a) Per-timepoint H_0 (top) and H_1 (bottom) persistence barcodes for the GSE134839 discovery cohort; permutation tests' closest approaches to significance are D9 ($p = 0.098$) and D2 ($p = 0.126$). (b) Pairwise Wasserstein distance heatmap between per-timepoint H_1 persistence diagrams: D0 is separated from all drug-treated timepoints in persistence-diagram space. (c, d) H_1 barcodes recomputed with and without 39 Tirosh cell-cycle genes ablated from the HVG set; the persistence-ratio ≥ 0.71 at every timepoint indicates loops are not cell-cycle artifacts.



(a) Max H_1 vs. days for the PC9 erlotinib and osimertinib time courses; osimertinib shows a $2.7\times$ monotonic increase from D0 to D14.



(b) Persistence barcodes for the YU-003 and YU-006 PDX models under untreated and osimertinib-residual conditions; both PDX models increase H_1 persistence under residual disease.

Figure 4: Drug-validation cohorts: osimertinib in PC9 and in EGFR-mutant patient-derived xenografts. (a) PC9 max- H_1 trajectory under erlotinib vs. osimertinib (GSE150949). (b) PDX YU-003 and YU-006 untreated vs. residual barcodes (GSE243562). Together these are the within-system monotonic-increase replications behind the cross-cohort summary in fig. 5.

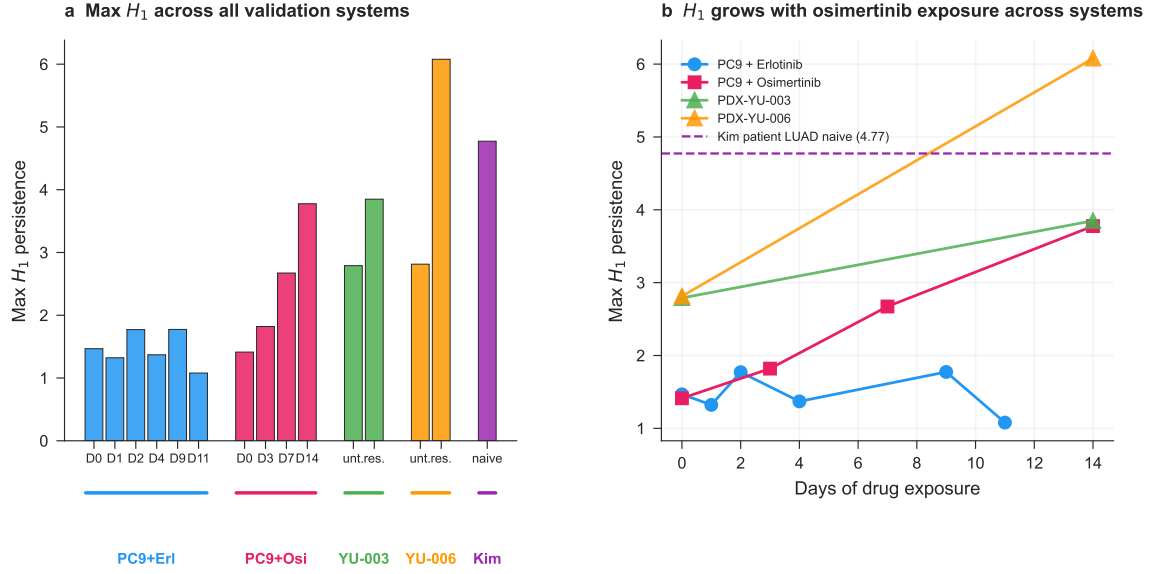


Figure 5: **Master comparison across all biological systems.** (a) Max H_1 bar chart across all conditions. (b) Max H_1 vs. days for all drug-treated systems, with patient-naive baseline as a reference line. Drug treatment pushes cell line and PDX topology toward patient-tumor-like complexity.

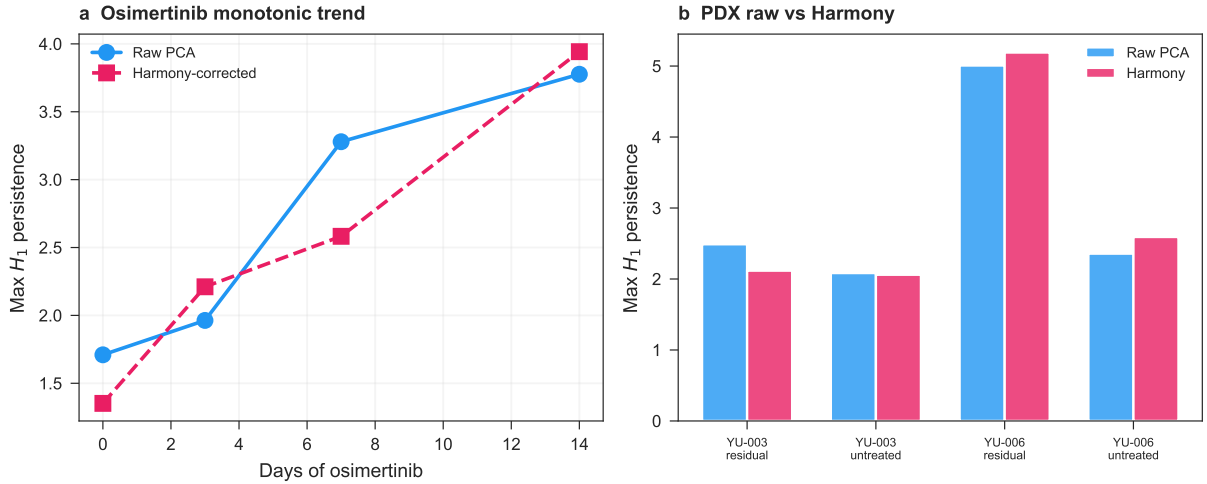
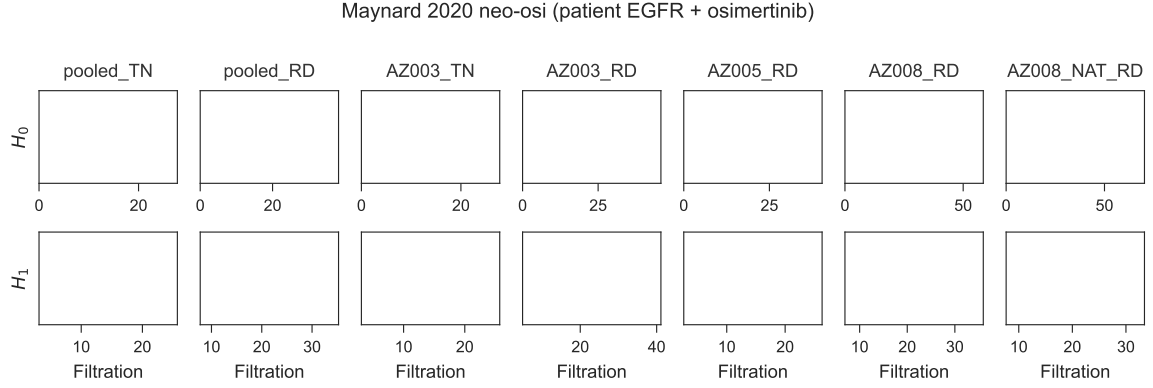
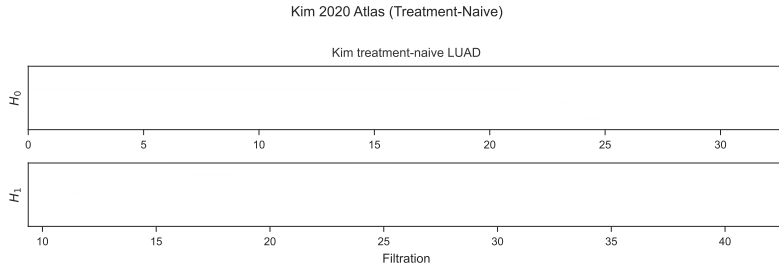


Figure 6: **Harmony batch correction sensitivity.** (a) Max H_1 vs. days of osimertinib exposure in the PC9 cell line cohort (GSE150949), computed on raw PCA (blue) and on Harmony-corrected embeddings (pink, dashed). Both show monotonic growth from D0 to D14. (b) Max H_1 per PDX group (untreated vs. residual disease) in GSE243562, raw (blue) vs. Harmony-corrected (pink). The YU-006 pre/post effect is preserved after batch correction.



(a) Maynard 2020 EGFR-mutant longitudinal cohort (14 patients, 13,244 cells): persistence barcodes (H_0 top, H_1 bottom) pooled across treatment-naïve (TN), persister (PER), and progressive disease (PD) stages. Pooled max H_1 : $5.37 \rightarrow 7.13 \rightarrow 8.05$; three of three matched pre/post biopsies increase post-treatment (median $1.60\times$).



(b) Treatment-naïve lung adenocarcinoma reference from the Kim et al. 2020 atlas (GSE131907, $n = 1,500$ epithelial cells subsampled from 5,000 tLung cells): persistence barcodes establishing the patient-tumor max $H_1 = 4.77$ reference used in fig. 5.

Figure 7: Patient-cohort validation across two independent scRNA-seq studies. (a) Maynard 2020 14-patient EGFR-mutant longitudinal cohort: max H_1 rises monotonically TN $\rightarrow$ PER $\rightarrow$ PD with 3/3 matched pre/post pairs increasing. (b) Kim 2020 LUAD atlas: persistence barcodes for the treatment-naïve primary-tumor epithelium reference. See Supplementary Table S11 for full per-patient Maynard trajectories.

Table 1: **Per-timepoint H_1 summary (GSE134839, erlotinib).** Permutation p -values from 500 within-cell gene-label shuffles.

Timepoint	n cells	Max H_1	# H_1 features	Perm. p (500)
D0	1560	1.47	1393	0.394
D1	324	1.32	255	0.552
D2	230	1.77	124	0.126
D4	200	1.37	54	0.144
D9	379	1.77	204	0.098
D11	249	1.08	124	0.266

Table 2: **Validation cohort H_1 permutation tests (100 permutations, 1,000-cell subsamples).** Values from 1,500-cell subsample used in main analysis; permutation tests used 1,000-cell subsamples (see Methods). Per-cohort max H_1 values vary across subsample sizes (1,000 vs. 1,500 cells) due to the strong subsample dependence of persistent homology on cell count (Supplementary Table S7, CV 8.9–23.5%); values reported in Table 2 use the 1,000-cell permutation-test subsample for internal consistency with the p -value column. Note that PC9 osimertinib D0 max H_1 is reported as 1.41 in the main analysis (5,000-cell subsample) and 1.74 here (1,000-cell permutation subsample); these are different subsamples of the same underlying data and are not inconsistent.

Dataset	Group	Observed H_1	p -value
PC9 + Osimertinib	D0	1.74	0.030
PC9 + Osimertinib	D7	3.08	< 0.01
PC9 + Osimertinib	D14	3.78	< 0.01
PDX YU-006	Untreated	3.62	< 0.01
PDX YU-006	Residual	6.08	< 0.01
PDX YU-003	Residual	3.85	< 0.01

Table 3: **Cell-cycle ablation controls.** Top: Tirosh cell-cycle gene removal from HVG set (39 of 97 Tirosh genes present; full discovery-cohort timepoints from GSE134839). Ratio is max H_1 after / before ablation; ratios ≥ 0.71 indicate loops are not cell-cycle artifacts. Bottom: stringent `sc.pp.regress_out` of S-score and G2M-score on 1,000-cell subsamples across five cohorts. Ratio is max H_1 after regression / original; every ratio > 1 indicates the H_1 signal is preserved after removing the dominant cell-cycle variance axis.

Group	n cells	H_1 before	H_1 after	Ratio
<i>Tirosh cell-cycle gene ablation (GSE134839 erlotinib)</i>				
D0	1560	1.47	1.35	0.92
D1	324	1.32	1.28	0.97
D2	230	1.77	1.57	0.89
D4	200	1.37	0.97	0.71
D9	379	1.77	1.39	0.78
D11	249	1.08	1.20	1.11
<i>Stringent S/G2M regression (1,000-cell subsamples)</i>				
Erl D0	1000	1.27	2.85	2.25
Erl D9	379	1.77	1.89	1.07
Erl D11	249	1.08	4.37	4.05
PDX YU-006 untreated	1000	2.89	3.67	1.27
PDX YU-006 residual	1000	4.65	5.96	1.28

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